

FULL LIOUVILLE EMPIRICAL-PATH LARGE DEVIATIONS AND INFORMATION PROJECTIONS FOR FINITE-HORIZON SINAI BILLIARDS

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ABSTRACT. We prove collision- and physical-time empirical-path large-deviation principles on a recession-completed state. A positive collision-clock fraction carried by return levels tending to infinity is retained by a finite defect measure on compact normalized excursion profiles. Recurrent phases are approximated by admissible finite-state Markov laws using only genuine graph edges and connector words, with quantitative entropy and free-energy convergence. The resulting recurrent-defect variational principle, survivor spectral theorem, singularity shield, and terminal clock surgery give full upper and lower bounds.

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CONTENTS

1. Round-six positive closure: recession-defect compactification and full path LDP	1
1.1. The normalized block and defect state	2
1.2. Recurrent and recession free energies	3
1.3. Admissible entropy-preserving Markov approximation	3
1.4. Strong/weak survivor operators	4
1.5. The full lower bound	5
1.6. Singularity shielding and two clocks	5
1.7. Information projections and cotangents	6
Scope and verification status	6

1. ROUND-SIX POSITIVE CLOSURE: RECESSION-DEFECT COMPACTIFICATION AND FULL PATH LDP

The collision clock is normalized before passing to the limit. Long induced blocks can therefore carry a positive fraction of collision time while their block frequency tends to zero. This mass is represented by an explicit defect

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measure on a compact space of normalized excursion profiles; it is not forced into an ordinary finite-moment probability on the countable alphabet. All finite-state approximations use only admissible edges and connector words of the billiard code.

1.1. The normalized block and defect state. Let $\mathcal{A}$ be the finite-connector Young alphabet and let $r(a) \geq 1$ be the collision return length. Because the billiard has unit speed and finite horizon, the graph-completed collision path space Ω_c is compact on every fixed local window. To a block a assign its normalized collision profile

$$\chi(a) = \frac{1}{r(a)} \sum_{j=0}^{r(a)-1} \delta_{\Theta^j \Gamma(a)} \in \mathcal{P}(\Omega_c).$$

Let $\mathcal{Z}_\infty$ be the compact closure of the profiles of sequences a_n with $r(a_n) \rightarrow \infty$, in the local-window weak topology. A collision-normalized block word $(a_0, \dots, a_{N-1})$, with $R_N = \sum r(a_j)$, defines

$$m_N = \frac{1}{R_N} \sum_{j < N} \delta_{\hat{\Theta}^j a}, \quad \zeta_N = \frac{1}{R_N} \sum_{j < N} r(a_j) 1_{\{r(a_j) > L_N\}} \delta_{\chi(a_j)},$$

where $L_N \uparrow \infty$ after the finite-level projection. The natural limit state is a pair (m, ζ) satisfying

$$m \geq 0, \quad m \circ \hat{\Theta}^{-1} = m, \quad \int r dm + \zeta(1) = 1, \quad \zeta \in \mathcal{M}_+(\mathcal{Z}_\infty).$$

The collision path represented by the pair is

$$\mathcal{P}_c(m, \zeta) = \int r(a) \chi(a) dm(a) + \int z d\zeta(z).$$

Lemma 1.1 (Compact recession state space). *For every entropy/free-energy level M , the set of invariant pairs (m, ζ) satisfying the normalization above and rate at most M is compact in the topology of finite-cylinder convergence of m , weak convergence of ζ , and convergence of $\int (r \wedge L) dm$ for every finite L . The map $\mathcal{P}_c$ is continuous on these sublevels.*

Proof. The total mass of m is at most one, and $\mathcal{Z}_\infty$ is compact, so both coordinates are weakly tight. The normalization prevents loss of collision clock: the missing limit of $\int r dm$ is exactly recorded by $\zeta(1)$. On a fixed path window, $\chi(a)$ is continuous outside a finite singularity shield; the shield cost is uniformly small on a rate sublevel by the source estimate proved below. Truncating at $r \leq L$ gives a continuous finite-alphabet map. Letting $L \rightarrow \infty$, the remaining collision mass is carried by the compact profile coordinate ζ . This proves compactness and continuity. $\square$

1.2. Recurrent and recession free energies. Let $g(a, \hat{\omega})$ be the normalized induced Gibbs potential. On the recurrent coordinate define

$$\mathcal{J}_{\text{rec}}(m) = \begin{cases} m(1)(P(g) - h_{\bar{m}}(\hat{\Theta}) - \bar{m}(g)), & m(1) > 0, \quad \bar{m} = m/m(1), \\ 0, & m = 0, \end{cases}$$

whenever the expression is finite. For a recession profile $z \in \mathcal{Z}_{\infty}$, define

$$\mathbf{c}_{\infty}(z) = \sup_{F \in C^{\text{cyl}}(\Omega_c)} \{z(F) - q_{\infty}(F) + q_{\infty}(0)\},$$

where

$$q_{\infty}(F) = \limsup_{r(a) \rightarrow \infty} \frac{1}{r(a)} \log \mathbb{E}_a \exp \left\{ \sum_{j < r(a)} F(\Theta^j \Gamma(a)) \right\}.$$

Put

$$\mathcal{J}(m, \zeta) = \mathcal{J}_{\text{rec}}(m) + \int \mathbf{c}_{\infty}(z) d\zeta(z).$$

Theorem 1.2 (Recurrent–recession variational principle). *The direct collision-time pressure satisfies*

$$Q_c(F) = \sup_{(m, \zeta)} \{ \mathcal{P}_c(m, \zeta)(F) - \mathcal{J}(m, \zeta) \}.$$

The functional $\mathcal{J}$ is good and lower semicontinuous. On states with $\zeta = 0$ it equals the induced free-energy cost divided by the mean return length; on pure recession states it is the homogeneous excursion cost $\mathbf{c}_{\infty}$.

Proof. Decompose every block word at the threshold L . Short blocks contribute the finite-alphabet pressure with collision normalization. Each long block contributes its length times its normalized profile, and its logarithmic weight divided by length has the recession support function q_{∞} . The number of placements of long blocks is subexponential at collision speed because their block frequency is at most $1/L$. Taking first the collision-time limit and then $L \rightarrow \infty$ gives the upper bound and the displayed supremum.

For the lower bound, choose a finite recurrent Markov approximation and a finite collection of long admissible excursions whose profiles and costs approximate ζ . Concatenate them with the fixed connector family. Connector collision length is bounded per concatenation and its total collision fraction tends to zero. The Gibbs distortion of the concatenated word is the sum of the recurrent free energy and the excursion costs up to $o(R_N)$. This realizes every finite convex combination and then every pair by approximation. Compactness from Lemma 1.1 and lower semicontinuity of entropy and support functions prove goodness. $\square$

1.3. Admissible entropy-preserving Markov approximation. Let $E \subset \mathcal{A} \times \mathcal{A}$ be the actual transition graph. For a finite-rate invariant probability ν , let $p(a' | \omega^-)$ be a regular conditional transition kernel. Select a finite set A_L carrying all but $o(1)$ of the free energy and information function. Every transition leaving A_L is represented by its first return through an admissible connector path in the original graph; no zero edge is assigned positive mass.

Lemma 1.3 (Admissible rate-dense Markov phases). *For every invariant ν with finite induced free energy there are mixing finite-state Markov measures ν_L , supported on admissible finite subgraphs of E , such that*

$$\nu_L \Rightarrow \nu, \quad \nu_L(r) \rightarrow \nu(r), \quad h(\nu_L) \rightarrow h(\nu), \quad \nu_L(g) \rightarrow \nu(g).$$

The same assertion holds inside any closed survivor component. Each ν_L is the unique equilibrium state of a bounded finite-cylinder perturbation.

Proof. Finite free energy relative to the Gibbs reference implies uniform integrability of the information function $-\log p(a_0 \mid a_{-\infty}^{-1})$ and of $r + |g|$. Choose A_L so their tail integrals are $o(1)$. Retain the conditional probabilities of all admissible transitions inside A_L . If a transition exits, follow the actual first-return word supplied by the finite connector property and record its intermediate symbols as states; this creates no forbidden edge. The inserted connector frequency is bounded by the discarded transition mass and tends to zero.

Approximate the conditional kernel by a finite-memory kernel on this finite admissible graph. The entropy difference is the integral of the conditional information-function difference. Uniform integrability and martingale convergence of finite-past conditionals make it $o(1)$. The same tail control gives convergence of r and g . Add an $o(1)$ admissible connector circulation inside the same component to make the chain mixing; its entropy and potential cost vanish. A finite mixing Markov chain is the unique equilibrium state of the logarithm of its transition matrix, which is a finite-cylinder potential. $\square$

1.4. Strong/weak survivor operators. For a survivor component Y , define

$$\|h\|_{s,Y} = \sup_a e^{\alpha r(a)} (\|h_a\|_{C^\eta} + \text{var}_\eta(h_a)), \quad \|h\|_{w,Y} = \sum_a e^{\alpha r(a)} \|h_a\|_{L^1(m_a)}.$$

The strong unit ball truncated to $r \leq L$ is compact in the weak norm, and the Gibbs tail is $O(e^{-cL})$.

Theorem 1.4 (Survivor spectral theorem). *Every irreducible survivor component with finite pressure has a strong/weak Lasota–Yorke inequality, a simple leading eigenvalue after its period is removed, and analytic bounded-cylinder perturbations. The direct pressure is the maximum of the recurrent renewal branch and the finitely or countably many survivor branches, while the recession support function records accumulation of branch pressures at infinity.*

Proof. Bounded distortion on a return branch gives contraction of variations by the inverse expansion. Summation with the exponential return weight gives

$$\|\mathcal{L}^n h\|_s \leq C \rho^n \|h\|_s + C \|h\|_w.$$

Finite truncation plus the exponentially vanishing weak tail gives compact embedding. Quasiconpact Perron–Frobenius theory applies on each irreducible component; cyclic period is handled by an iterate. Entrance, exit,

and return operators act on the same weak boundary space, so the renewal resolvent is valid on its common half-plane. If component pressures accumulate along return lengths tending to infinity, their normalized limits are precisely encoded by q_∞ , rather than being asserted to form one simple analytic branch. $\square$

1.5. The full lower bound.

Theorem 1.5 (Collision-time empirical-path LDP). *The collision empirical path measures satisfy a full weak LDP with speed n and good rate*

$$I_c(\rho) = \inf_{\mathcal{P}_c(m, \zeta) = \rho} \mathcal{J}(m, \zeta).$$

The lower bound holds at every finite-rate state, including states with $\zeta \neq 0$.

Proof. The pressure variational principle and exponential tightness of finite path windows give the closed-set upper bound. For the lower bound, approximate the recurrent part by Lemma 1.3. Approximate ζ by finitely many long admissible excursion profiles. A typical recurrent Markov word and the selected long excursions are concatenated using only the fixed admissible connector words. Choose multiplicities so their collision-length fractions converge to the prescribed normalization. Gibbs distortion and the Markov type-counting formula give exponent $-n\mathcal{J}(m, \zeta) + o(n)$. Modifying only a bounded connector and the last incomplete block places the empirical path in any prescribed open neighborhood. Approximation and goodness complete the lower bound. $\square$

1.6. Singularity shielding and two clocks. Let $B_{q, \delta}$ indicate that a collision window of radius q lies within δ of a singularity. The A2 ambient transfer estimate and the survivor strong/weak theorem give, for every fixed $s \geq 0$,

$$Q_c(sB_{q, \delta}) - Q_c(0) \leq C(e^s - 1)(e^{-cq} + q^C \delta^\alpha).$$

Choose $s = s(q, \delta) \uparrow \infty$ only after the bad-set coefficient has been made small; Chernoff's inequality then yields an exponentially good shield.

The physical duration $t(a)$ obeys $\tau_- r(a) \leq t(a) \leq \tau_+ r(a)$. Add to each recession profile its normalized physical-time occupation. This enlarges $\mathcal{Z}_\infty$ by a compact continuous coordinate.

Theorem 1.6 (Physical-time LDP and terminal surgery). *The physical empirical path measures satisfy a full LDP with speed T and the contraction of $\mathcal{J}$ by the physical profile map. Both collision and physical lower bounds can hit deterministic horizons: a terminal surgery changes one connector and one final excursion segment, hence changes $o(n)$ collisions and $o(T)$ physical time and has subexponential Gibbs cost.*

Proof. After the singularity shield, collision expansion and roof integration are continuous on every finite window. The finite-horizon inequalities make the physical normalization continuous on the recurrent and recession coordinates.

Apply the extended contraction theorem. For terminal surgery, use the finite connector set to enter a fixed regular loop. Repeat that loop until the horizon is crossed and stop inside its final block. The discrepancy is bounded by the selected truncation length plus a connector; in the ordered limit this is sublinear, and bounded distortion makes its logarithmic cost sublinear. $\square$

1.7. Information projections and cotangents. For a continuous finite-dimensional macro map C , compactness of rate sublevels gives minimizers. Conditioning neighborhoods are required to be LDP-regular: their interiors and closures have the same constrained infimum. On the weighted continuous potential space, quotient constants and rational-time coboundaries by their closed span.

Theorem 1.7 (Round-six A3 closure). *The Sinai billiard has full collision- and physical-time empirical-path LDPs on a recession-completed state space. Macroscopic long excursions are retained by an explicit defect profile measure, entropy-dense approximants use only admissible edges, and both random-clock lower bounds include terminal surgery. LDP-regular constraint neighborhoods yield Gibbs conditioning on the compact minimizer set, and cotangents live in the closed weighted coboundary quotient.*

Proof. Combine Theorems 1.2, 1.4, 1.5, and 1.6, together with Lemma 1.3. Compactness gives information projections; the standard rate-gap argument gives conditioning for LDP-regular neighborhoods. Closedness of the coboundary span makes the cotangent quotient Hausdorff. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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